ARM-450 — Why the Design Changed

From a printed arm that failed in service, to ARM-450, and how both relate to a space-grade manipulator.

Revision 2 · 2026-09-02 · every figure derived in the supporting documents listed in §7

1. What this report answers

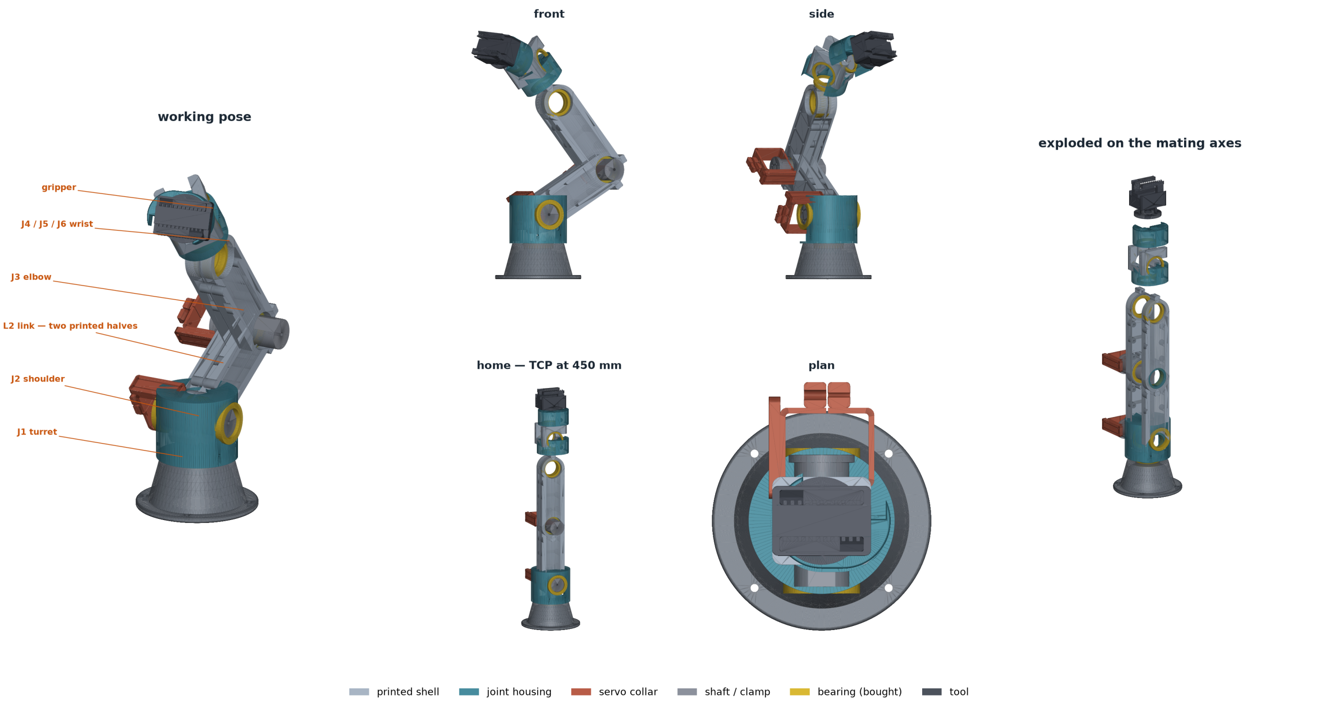
Three questions, in order:

1. What was the previous design, and what did it actually do?
2. Why was it changed, and what did the change buy?
3. How do the old arm, ARM-450 and a real space manipulator compare — where they are the same, and where they are not?

Everything here is measured or derived. Where a number is an assumption rather than a measurement, it says so.

ARM-450 · detailed assembly · 6-DOF · 450 mm to the tool face

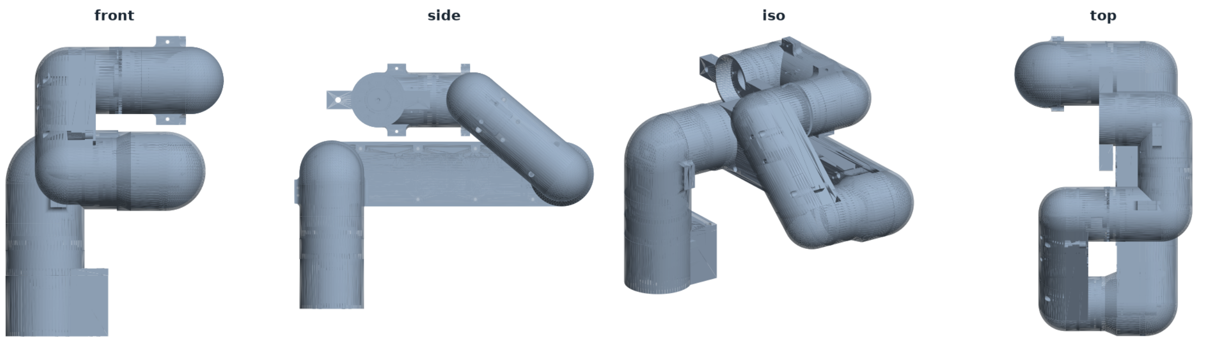
32 components from arm450_assembly_*.step — printed structure, bearings, joint shafts, servo collars and the quick-change gripper



2. The previous design

A printed 6-DOF arm, exported from Fusion 360, built and run. It broke. Three faults were diagnosed, and all three were measured from the geometry rather than guessed at from the symptoms.

Existing design — Robo_2_assembled.stl (envelope 155 × 260 × 213 mm, as-saved pose)



2.1 The servo mounts snapped

The mounts were sized for the torque the arm carries in normal motion — 2.08 N·m at the shoulder. But a servo does not deliver its operating torque into its own mount. In a collision, on a commanded step, or when a joint hits its limit, it delivers stall torque: 2.94 N·m for an ST3215, 4.90 N·m for an ST3250.

The 4 mm bracket runs at 17–95 MPa depending on where it grips, against a realistically sustainable ~9 MPa for printed PLA under repeated load. It fails under every assumption in that range, and it did.

There is a second trap inside the first: fitting a stronger ST3250 multiplies mount stress 2.36× while changing the load the arm actually carries by nothing at all. A bigger motor makes the structural problem worse.

2.2 The joints wobbled, and tuning could not fix it

Each joint ran on a Ø42 × 3.7 mm thrust ball bearing. A thrust bearing carries axial load — its balls run between two flat races, so the joint is free to tilt. A robot joint's dominant load is a bending moment: 2.08 N·m at the shoulder from everything outboard on a long lever.

Angular play follows $\theta = 2c/d$. With effectively no bearing spacing ($d \approx 5$ mm) and 0.2 mm clearance, that is 28.9 mm of wobble at 360 mm reach — larger than every other error source in the arm combined.

The oscillation that came with it was mechanical, not a control problem. Clearance in a position-controlled joint is a dead zone: the servo turns and nothing moves until the slack takes up, then it moves suddenly. A feedback loop pushing against a dead zone produces a limit cycle. Lowering the gain makes the arm sluggish and the oscillation smaller — never absent.

2.3 A revision opened a hole in the workspace

Revision 1.1 lengthened one link and not the other: 195.1 mm joint-to-joint against 97.9 mm. The reachable region of a two-link chain is an annulus whose inner radius is |L2 −

L3|, so that split created a Ø194 mm dead zone directly in front of the robot — and the chain totalled 453 mm, over the 450 mm limit it was meant to meet.

The original links were 97.9 / 97.9: equal, and with no dead zone at all. The revision broke something that had been right.

3. Why the shift

The three faults are not unrelated. Each is the same kind of error — a design sized against the wrong quantity — and each has a general rule behind it.

the fault	what it was sized against	what it should have been sized against
Mounts snapped	the torque the arm uses	the torque the servo can deliver
Joints wobbled	the shaft diameter	the direction of the load
Dead zone	one link at a time	the difference between the links

That is the shift. Not "make it stronger" — the previous arm was not weak in the places it broke, it was strong against the wrong load. ARM-450 changes what each part is designed against:

> Wherever an actuator can drive its own structure to failure, the actuator's limit is the design load. Gravity torque sets what the servo must do; stall torque sets what the mount must survive.

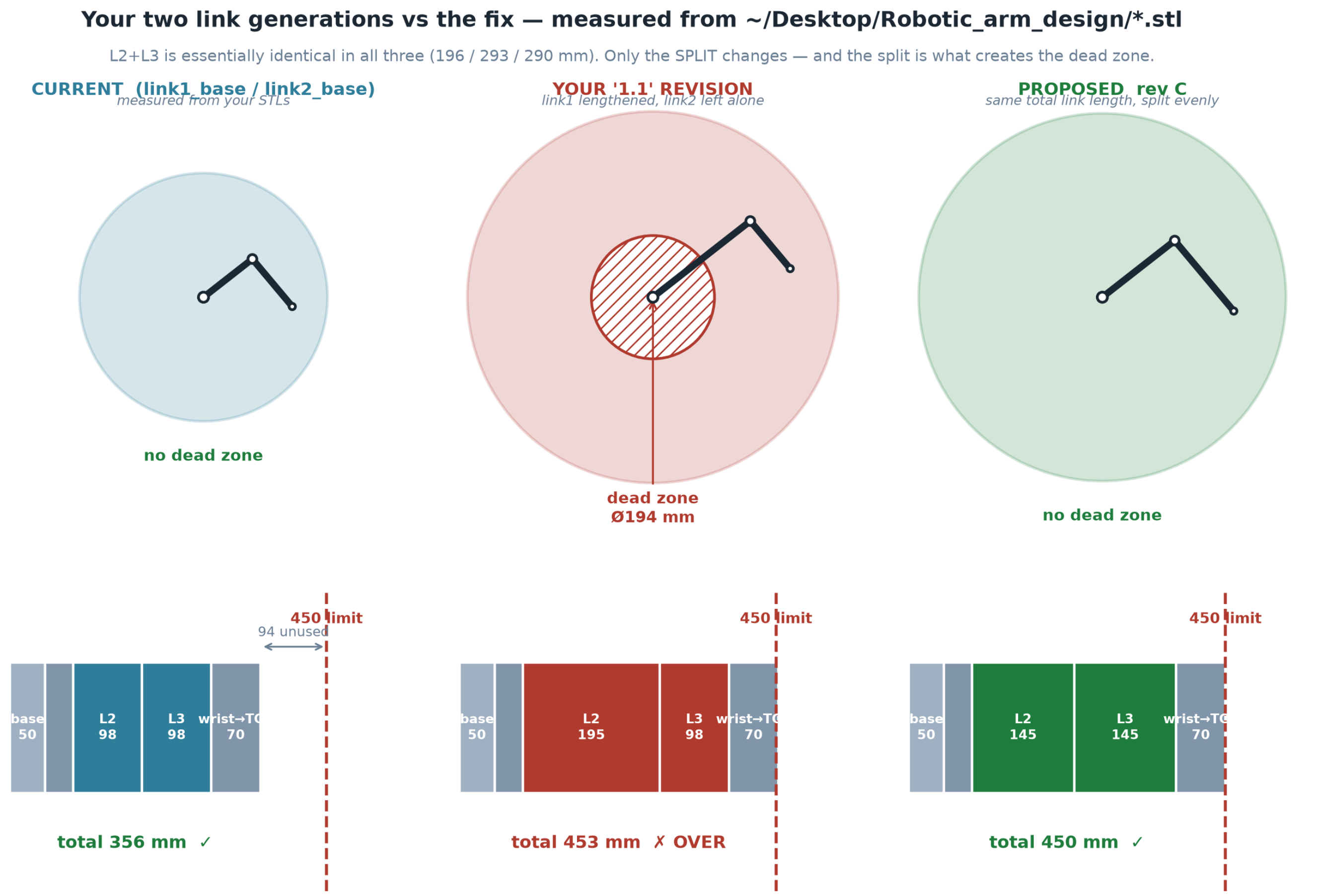
> Match the bearing to the load direction, not to the shaft diameter. Moment stiffness scales with bearing spacing squared — worth far more than bearing size or count.

> For a two-link arm, L2 = L3 eliminates the dead zone, and it is nearly free. Shoulder torque is almost flat in the split, because the dominant term is the payload at a fixed reach.

4. What changed, and what it bought

item	previous	ARM-450	benefit
Mount sizing load	2.08 N·m operating	stall, 2.94 / 4.90 N·m	the failure mode that broke the arm is designed out
Joint bearing	Ø42 × 3.7 thrust washer	6806 pair, 40 mm apart	wobble 28.9 mm → 0.65 mm
Wrist bearing	as above	6706 pair	−92.6 g, no loss of moment stiffness
Joint preload	none	M5 × 90 through-bolt	removes the dead zone that caused the limit cycle
Link lengths	195.1 / 97.9 mm	119.0 / 119.0 mm	Ø194 mm dead zone eliminated

item	previous	ARM-450	benefit
Chain total	453 mm	450.0 mm	inside budget
Bending axis	about the section width	about its depth	load is transverse to the link
Section depth	29.0 mm	50.0 mm at the boss	stiffness ∝ depth³
Link construction	single shell	split shell, tongue-and-groove seam	prints flat, no supports, no bridging over bores
Open-section penalty	applied	withdrawn	seam bolts close the section
Backlash budget	J2, J3 only	all six joints	the error budget is now complete
Tool interface	bolted	three-lug bayonet, 30° twist	tool change without three blind M3s under a wrist



4.1 The benefits, stated as results

	previous	ARM-450
Joint wobble at full reach	28.9 mm	0.65 mm

	previous	**ARM-450**
Workspace dead zone	Ø194 mm	none
Mount stress vs allowable	17–95 MPa vs ~9 MPa — fails	worst p99 safety factor 23.0
Tip deflection under load	not analysed	0.197 mm against a 0.30 mm limit
Chain length	453 mm (over)	450.0 mm
Vertical sine trace	not achievable	0.014 mm rms, 240/240 waypoints clear
Horizontal reach	—	360 mm (myCobot 280: 300 mm)
Vertical surface drawable	—	52.3 % (myCobot 280: 57.9 %)

The kinematics were derived by hand as a modified (Craig) DH table and then checked against two independent routes — the URDF chain and ikpy, an outside package. Agreement is 2.17×10^{-13} mm. Three routes cannot share one mistake.

4.2 The honest limit

The structure is no longer what limits accuracy, and has not been for some time. Backlash in the hobby serial servos accounts for about 98 % of the predicted tool error — 2.25 mm of a 2.25 mm total, against the structure's 0.197 mm. It is also the one significant quantity in this report that has not been measured. Building the arm tests that assumption; it does not primarily test §4.1, which has been checked three ways.

5. Old arm, ARM-450, and a space-grade manipulator

Sixteen subsystems were compared against a flight equivalent. Three survive unchanged, two need rework, eleven must be replaced.

What survives a move to flight hardware, and what does not

SUBSYSTEM	ARM-450 (this build)	SPACE-GRADE EQUIVALENT
Kinematics & DOF layout	6R, 450 mm, offset-free wrist	identical
Joint architecture	Ø42 pocket · 6806 · Ø30 shaft, one part number throughout	same scheme, flight bearings
Fastening scheme	M3 steel, heat-set brass inserts	same layout; locked, staked, vented
Structural material	plain PLA, E = 3.5 GPa, Tg 60 °C	Al 6061-T6 or CFRP, E 70 GPa
Mass	1.584 kg assembled (1.34 structure + 0.24 fasteners)	≈1.13 kg in Al 6061, ~10× stiffer
Actuators	ST3215/3250 hobby serial servos, plastic gear train	BLDC + harmonic drive, vacuum-rated, rad-tolerant
Feedback	servo-internal pot, no joint encoder	joint-side resolver / optical encoder
Precision	≈2.2 mm, backlash-dominated	< 50 µm, encoder-closed
Lubrication	wet lithium grease in 6806	dry film MoS_2 / PFPE, vacuum-stable
Thermal range	0–50 °C; Tg 60 °C is the wall	−120 to +120 °C cycling, CTE-matched
Vacuum compatibility	none — PLA outgasses, trapped volumes unvented	TML < 1 %, CVCM < 0.1 %, vented
Radiation	none	TID-rated parts, latch-up immune
Qualification	analysis + bench only	vibration, shock, thermal-vac, EMC
Redundancy	single string	dual-wound motors, redundant sensing
Docking interface	3-lug bayonet, J6-driven, ±9.75 mm capture	same principle; softer capture, latch preload, fluid seal
Cost	consumer printer + hobby servos	3–4 orders of magnitude higher

survives the transition unchanged · survives in principle, needs rework · does not survive — must be replaced

5.1 Where all three are the same

	why it transfers
6R serial kinematics	A six-revolute chain with this layout is unaffected by what it is made of. The URDF, the IK and the workspace transfer directly from bench to flight.
One bore, one shaft, one bearing family per joint	Good practice at any grade. Only the part numbers and the lubricant change.
Bolt patterns and load paths	Flight versions add locking compound, staking and vented screws — in the same places.

The old arm shares the first of these and fails the second: it had the serial architecture right and the joint architecture wrong.

5.2 Where ARM-450 and space-grade diverge

	ground manipulator	**space manipulator**
Gravity	dominates sizing; the arm must hold itself up	~0 g. Sizing is driven by inertia and dynamics
What limits payload	actuator torque against gravity	reaction on the base — the arm moves the spacecraft

	ground manipulator	space manipulator
Base	bolted to a rigid floor	free-floating; momentum must be managed
Structure	stiff enough not to sag	stiff enough not to oscillate — no damping in vacuum
Thermal	20 ± 15 °C	−120 to +120 °C; CTE mismatch drives joint clearances
Lubrication	wet grease	dry film (MoS_2) or PFPE — wet grease outgasses and creeps
Materials	anything	TML < 1 %, CVCM < 0.1 %; no trapped volumes
Radiation	none	TID-rated electronics, latch-up immunity
Backlash	tolerable	often unacceptable — no operator in the loop
Repair	trivial	none. Single-fault tolerance or redundancy
Qualification	"it works"	vibration, shock, thermal-vac, EMC, life test
Control	position	often impedance or force — contact dynamics with a free target

The four that rule ARM-450 out as a flight article:

1. The material is the wall. PLA's glass transition is ~60 °C; a sun-facing surface in LEO exceeds it, and it outgasses. Re-made in Al 6061-T6 the same geometry is ≈1.13 kg — lighter than printed — and about 10× stiffer. The swap is an improvement that costs money, not a compromise.
2. The actuators. Hobby serial servos: plastic gears, wet grease, no radiation tolerance, no joint-side feedback. This one choice sets the 2.2 mm precision figure.
3. Vacuum compatibility was never designed in. The printed parts contain closed internal volumes with no vent path.
4. Nothing has been qualified.

5.3 What genuinely is space-relevant

The docking interface. Capture tolerance is solved as geometry, not as accuracy: a Ø34 mouth over a Ø16 throat gives a measured ±9.75 mm capture envelope against ≈2.2 mm of arm error. That is exactly how real docking hardware works, and it is the transferable idea — you do not make the arm accurate enough to hit a hole, you make the hole forgiving enough to be hit.

Fluid transfer. A radial O-ring in the spigot, compressed by the bayonet itself, with a Ø8 pass-through in both halves: the architecture of an orbital refuelling coupling at demonstration scale.

No seventh actuator. The docking latch is driven by J6, which has ±175° of travel and nothing to do during a docking approach. Using an existing degree of freedom to drive a mechanism is a mass-saving technique that matters far more in orbit than on a bench.

The verification method, which is grade-independent: every dimensional check reads the exported geometry and measures what is actually there, rather than trusting what the parameter file says should be there.

> The honest framing. ARM-450 is a kinematic and architectural prototype of a space manipulator and a functional demonstrator of a docking interface. It is not a flight article, and it is not presented as one. What transfers is the geometry, the joint scheme, the docking principle and the verification method. Eleven of sixteen subsystems do not transfer, and §5.2 names them.

6. Conclusion

The previous arm failed for three reasons, and all three were design errors rather than manufacturing faults: mounts sized against the load the arm uses instead of the load the servo can deliver, a bearing chosen for the wrong load direction, and a link revision that opened a dead zone in a workspace that had been correct. Each had a measurable signature — 17–95 MPa against a 9 MPa allowable, 28.9 mm of wobble, a Ø194 mm unreachable region.

The shift was not to make the arm stronger. It was to change what each part is designed against. Every load case now runs at servo stall; the joints run on spaced, preloaded bearing pairs chosen for moment capacity; the links are equal.

What that bought is measurable: wobble from 28.9 mm to 0.65 mm, the dead zone gone, a worst-case structural safety factor of 23, 0.197 mm of tip deflection against a 0.30 mm limit, and a vertical sine traced at 0.014 mm rms with every waypoint clear. The arm is 450.0 mm to the tool face, prints in plain PLA on a 0.4 mm nozzle in about 21 hours from thirteen parts with no supports, and its kinematics agree with two independent implementations to 10^{-13} mm.

Against a space-grade manipulator the picture is deliberately unflattering and deliberately specific: three subsystems transfer, eleven do not, and the reasons are material, actuator, vacuum compatibility and qualification — not geometry. The geometry is the part that would survive, and the docking interface is the part worth carrying forward, because it solves capture the way flight hardware solves it: with tolerance built into the shape rather than demanded of the arm.

One item remains open. The assembled arm is 1584 g against a 1450 g ceiling — 134 g over, and the overage is fasteners: 244 g of inserts, through-bolts and screws that were absent from the mass model until it was made a hard check. It is reported rather than absorbed. Either the figure is accepted and recorded, or the base is lightened and its FEA re-run.

7. Supporting documents

document	what it carries
ARM450_LESSONS.pdf	every error found in the previous arm, with the reasoning that exposed it
ARM450_REPORT.pdf	the full engineering record, part by part
ARM450_DEFENCE.pdf	review dossier, method, and anticipated questions with answers
ARM450_KINEMATICS.pdf	DH derivation, Jacobian, IK and the ikpy cross-check
ARM450_BUY.pdf	bill of materials and print order
ARM450_DRAWINGS_ISO.pdf	per-part drawings